E207: LAB

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This report presents the experimental results of the characterization of the LAB electron beamline at the University of Bonn. The study focused on the calibration of corrector magnets, the centering of the 25 keV electron beam using beam-based alignment (BBA), and the measurement of the transverse beam emittance. Calibration of the corrector magnets revealed statistically significant systematic underreporting of kick angles by the control system, with calibration factors ranging from 1.14(2) to 1.55(5). Using BBA, the beam was centered in all four quadrupole magnets to decouple steering from focusing effects. Transverse emittance was determined via the quadrupole scan technique and the multi-screen method. The quadrupole scan yielded emittance values of $\epsilon_x = 4.13(5)\,\mathrm{mm}\cdot\mathrm{mrad}$ and $\epsilon_z = 3.65(7)\,\mathrm{mm}\cdot\mathrm{mrad}$ using the determinant method. The multi-screen method initially resulted in poor model agreement ($\chi^2/\mathrm{ndof} \sim 70$–$83$). A leave-one-out analysis identified screen S0 as a systematic outlier. Discarding S0 yielded an exact three-screen solution with emittances of $\epsilon_x = 4.5(3)\,\mathrm{mm}\cdot\mathrm{mrad}$ and $\epsilon_z = 3.9(2)\,\mathrm{mm}\cdot\mathrm{mrad}$, reducing the discrepancy with the quadrupole scans to approximately 10 %.

I. INTRODUCTION

This report documents the experimental characterization of the LAB beamline, a 25 keV electron beam facility at the University of Bonn designed for hands-on training in accelerator physics and beam dynamics. The beamline consists of five fluorescence screens for beam diagnostics, four quadrupole focusing magnets, and four corrector deflection magnets distributed along a 3 m beam path [1]. The primary physics goals are to:

1. Measure and quantify the relationship between control-system command signals and actual beam deflection angles in the corrector magnets (kick calibration).

2. Center the electron beam through each quadrupole magnet using beam-based alignment (BBA), decoupling steering from focusing effects.

3. Determine the transverse beam emittance in both horizontal and vertical planes using the quadrupole scan technique and multi-screen method.

The calibration of corrector magnets is done since the control system provides only a nominal relationship between input current and output kick angle. In practice the actual kick deviates from the nominal value by up to 55 % as later discussed in Section III A. Determining the true calibration factor $C = \alpha_{\mathrm{actual}}/\alpha_{\mathrm{sys}}$ is a prerequisite for all downstream alignment and emittance measurements.

Beam-based alignment ensures that the electron trajectory passes through the magnetic center of each quadrupole, preventing unintended effects on the beam due to off-axis propagation. This is achieved by measuring the beam position difference Δu between two quadrupole settings as a function of corrector current, finding the zero crossing where the beam is centered.

Emittance (the phase-space volume occupied by the beam) is a conserved quantity (by Liouville's theorem) that characterizes beam quality and limits the performance of downstream applications. We measure emittance using two independent methods: the quadrupole scan technique, which varies quadrupole strength and measures beam size evolution, and the multi-screen method, which measures beam size at fixed quadrupole settings and back-propagates to the initial conditions. Comparison of the two methods provides a consistency check on the accelerator model and identifies potential systematic effects.

II. SETUP

All information about the setup is taken from [1].

Electrons are emitted by a thermionic electron gun, accelerated electrostatically to a kinetic energy of 25 keV (≈ 30.2 % of the speed of light), and transported through an evacuated beam pipe of approximately 3 m length. The vacuum is maintained by an ion getter pump close to the gun ($\sim 10^{-8}$ mbar) and two turbo-molecular pumps further downstream ($\sim 10^{-7}$ mbar).

A schematic of the beamline is shown in Fig. 1. The lattice can be grouped into four modules. The first module (gun, two solenoids `SOL1`/`SOL2`, corrector `C0` and screen `S0`) prepares and pre-focuses the beam. The four following modules are nominally identical and each consists of a screen $\mathtt{S}_i$, a quadrupole $\mathtt{Q}_i$ and a corrector

C$_i$ ($i = 1, 2, 3, 4$). The screens always sit behind the quadrupole they are associated with, so that the beam profile is measured after the focusing element. To see the beam shape going into a quadrupole we look at the previous screen which sits directly in front of the quadrupole. Element lengths and drift spaces relevant for the matrix-optics calculations performed in this report can be seen in Appendix A.

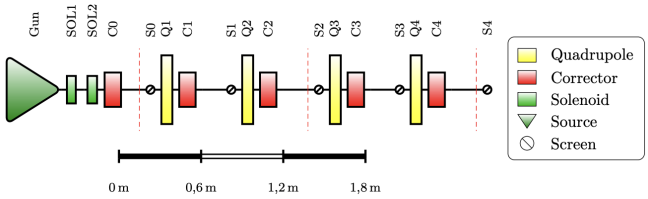


FIG. 1: Schematic drawing of the LAB beamline. The beam travels from left to right, starting at the electron gun, passing through two solenoids SOL1/SOL2, four corrector magnets C0–C4, four quadrupole magnets Q1–Q4 and five fluorescence screens S0–S4. Image taken from [1].

a. ***Magnets.*** Three magnet types are installed:

- **Solenoids** (SOL1, SOL2) directly behind the gun act as thick focusing lenses through their longitudinal field and fringe fields and provide the initial focusing of the divergent beam leaving the cathode.
- **Corrector magnets** (C0–C4) consist of two orthogonally mounted pairs of coils in an approximately Helmholtz configuration. They deflect the beam independently in the horizontal (x) and vertical (z) plane. The deflection angle in the small-angle approximation reads [1]

$$\alpha \approx \frac{eBL_C}{p}, \tag{1}$$

  with effective field length $L_C = 78.5\,\text{mm}$, magnetic field strength B, momentum p and elementary charge e. For ion-optical calculations, the corrector is treated as a pure drift, since the kick adds to the angle relative to the ideal beam line independent of position and angle of the incoming beam.
- **Quadrupole magnets** (Q1–Q4) have hyperbolically shaped iron pole shoes which produce a field that vanishes on the beam axis and grows linearly with the transverse displacement. Each quadrupole focuses in one transverse plane and defocuses in the orthogonal one. Their effective length is $L_Q = 74(1)\,\text{mm}$ and the half aperture is $a = 28\,\text{mm}$. The focusing strength k is [1]

$$k = \frac{e}{p}\frac{\partial B_z}{\partial x} = 128\,027(150)\,\frac{I}{p}, \tag{2}$$

  with $[k] = \text{m}^{-2}$, $[I] = \text{A}$ the respective magnet's current, $[p] = \text{keV}$ the momentum.

b. ***Beam diagnostics.*** The transverse beam position and spot size are measured with five screens S0–S4 as shown in Fig. 1. Each consists of an Al_2O_3 fluorescent plate tilted at 45° to the beam axis and imaged by a CCD camera mounted below the beam pipe. The analog video signal is digitised and analyzed by in-house frame grabber software, which fits a 2D Gaussian with a flat top to the intensity distribution and returns the beam centroid and the horizontal and vertical widths σ_x, σ_z. The 45° tilt makes the pixel-to-millimetre conversion position-dependent; this distortion is corrected internally by the framegrabber. Statistical fluctuations of the fit values are quantified by the RMS over the last ten images and are used as the position uncertainties throughout this report.

c. ***Control system.*** All magnet currents, corrector kicks and screen positions are operated through the LAB control system. Corrector deflections can be set either via a displayed kick angle α_{sys} or via the coil current I; the latter has a finer set-point resolution ($\sim 10^{-3}\,\text{A}$) and is therefore used for the beam-based alignment (Sec. III B). The relation between α_{sys} and the actual kick angle is not assumed to be unity but is determined experimentally in Sec. III A.

III. BEAM BASED ALIGNMENT

A. Corrector Kick Angle Calibration

For each corrector magnet (C0 and C1), the beam position on the downstream fluorescence screen (S0 and S1 respectively) was measured as a function of the displayed angle α_{sys}. With the quadrupole magnets off to ensure a linear trajectory, the beam deflection follows the geometry relation $\alpha_{\text{actual}} \approx x/d$, where x is the transverse displacement and d is the distance from the corrector effective-field center to the screen effective-center. The distances d were calculated from the lattice parameters in Appendix A :

- C0 → S0: $d = L_{\text{C0,eff}}/2 + \text{drift} + L_{\text{S0}}/2 = 0.0785\,\text{m}/2 + 0.185\,\text{m} + 0.04\,\text{m}/2 = 0.2442\,\text{m}$
- C1 → S1: $d = L_{\text{C1,eff}}/2 + \text{drift} + L_{\text{S1}}/2 = 0.0785\,\text{m}/2 + 0.316\,\text{m} + 0.04\,\text{m}/2 = 0.3753\,\text{m}$

where L_{eff} denotes effective field length and the drifts account for spacing between elements per the beamline schematic seen in Fig. 1. Seven measurement points were collected for each corrector in both horizontal (x) and vertical (z) planes, spanning kick angles from approximately $-18\,\text{mrad}$ to $-7.5\,\text{mrad}$ (C0) and similar ranges for C1. Uncertainties in the measured kick angles were 0.1 mrad (control system precision), and uncertainties in beam positions ranged from 0.01 mm to 0.021 mm (framegrabber fit uncertainties). A linear relationship $\beta = m \cdot \alpha_{\text{sys}} + b$ was fitted to each dataset using orthogonal distance regression (ODR), where α_{sys} is the

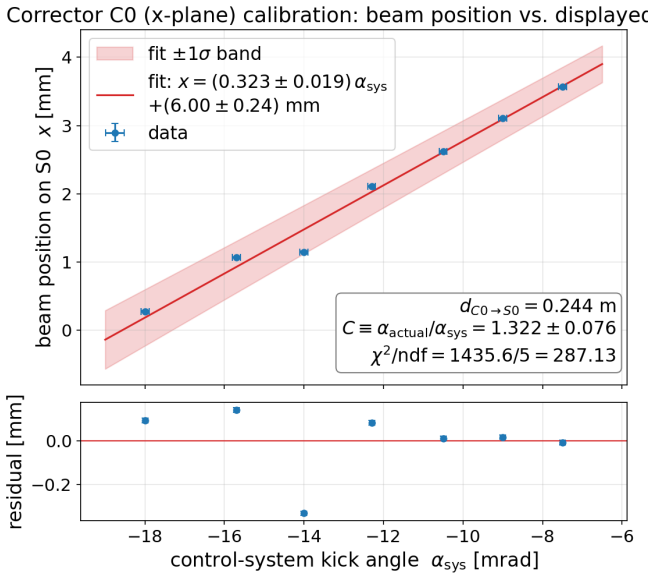


FIG. 2: Corrector C0 (x-plane) calibration: beam position on screen S0 as a function of displayed kick angle α_{sys}. The linear fit (red line with $\pm 1\sigma$ band) yields calibration factor 1.32(8) , indicating the control system underreports the actual kick by approximately 32%. The residual panel shows scatter ($\chi^2_{\mathrm{red}} = 287$), especially in one outlier at -14 mrad.

control-system displayed kick angle, β is the measured beam position in mm, m is the slope in mm/mrad, and b is the offset in mm. The fit band shown in the plots reflects the uncertainty in the fitted parameters (m and b) resulting from the measurement uncertainties. The calibration factor $C = \alpha_{\mathrm{actual}}/\alpha_{\mathrm{sys}} = m/d$ quantifies the relationship between displayed and true kick angles:

- C0 (x-plane): $C = 1.32(8)$, $\chi^2_{\mathrm{red}} = 287$ (poor fit quality)
- C0 (z-plane): $C = 1.55(5)$, $\chi^2_{\mathrm{red}} = 14.1$
- C1 (x-plane): $C = 1.14(2)$, $\chi^2_{\mathrm{red}} = 15.2$
- C1 (z-plane): $C = 1.20(1)$, $\chi^2_{\mathrm{red}} = 6.3$

All four correctors exhibit $C > 1$, indicating that the control system systematically *underreports* the actual kick by approximately 15–55%. This offset must be accounted for in subsequent alignment and beam diagnostics procedures. The z-planes show slightly larger factors than the x-planes for the same corrector.

During the initial analysis, the C0 x-plane fit showed an abnormally high $\chi^2_{\mathrm{red}} = 287$, driven entirely by a single measurement outlier at -14 mrad. Excluding this single anomalous point from the ODR fit yielded a corrected calibration factor of $C = 1.272(15)$ and restored the goodness-of-fit to $\chi^2_{\mathrm{red}} = 10.30$, bringing it in line with the other three datasets.

The reduced chi-squared values (χ^2/ndf from 6.3 to 15.2) indicate a poor statistical fit when considering only the framegrabber RMS uncertainties. However, the framegrabber RMS captures only the high-frequency temporal jitter of the beam at a static position. It does not account for spatial systematic errors introduced as the beam is steered across the screen, such as the worn out screens not uniformly measuring the intensity of the beam, which can shift the fitted centre of the beam. Consequently, the temporal RMS underestimates the total experimental scatter, but the linear model remains the correct physical description for extracting the calibration factors.

B. Aligning the Beam

To decouple beam steering from focusing, the beam was centered in each of the four quadrupoles Q1–Q4 using the beam-based alignment method described in Sec. 3.3 of [1]. For each quadrupole the corrector immediately upstream (C0 for Q1, C1 for Q2, C2 for Q3, C3 for Q4) was used to shift the beam, while the quadrupole current was toggled between two fixed values. The beam position on the screen directly behind the quadrupole was read at both quadrupole settings, and the difference

$$\Delta u(I) \;=\; u_1(I) - u_2(I), \qquad u \in \{x, z\}, \tag{3}$$

gives the displacement caused by the off-centre passage through the quadrupole. By Eq. (3) of [1] , this difference is, to first order, proportional to the offset of the beam from the magnetic centre, so the corrector setting at which $\Delta u = 0$ corresponds to the beam passing through the quadrupole centre. We worked with the corrector current I rather than the displayed kick angle α_{sys}, since I can be set with finer resolution.

For every quadrupole the procedure was carried out separately in the horizontal (x) and vertical (z) plane. Each quadrupole was aligned in both planes before moving on to the next downstream quadrupole to not destroy the alignment of the following quadrupoles.

Analysis

For each measurement, we fit a straight line $\Delta u(I) = mI + b$ to the data using Orthogonal Distance Regression (ODR), which accounts for uncertainties in both the corrector current I (set to last readable digit 0.001 A) and the position difference Δu. The position uncertainties are propagated from the framegrabber fit errors via $\sigma_{\Delta u} = \sqrt{\sigma_{u_1}^2 + \sigma_{u_2}^2}$. The zero crossing

$$I_0 \;=\; -\frac{b}{m}, \qquad \sigma_{I_0}^2 \;=\; \left(\frac{\sigma_b}{m}\right)^2 + \left(\frac{b\,\sigma_m}{m^2}\right)^2 - \frac{2b}{m^3}\,\mathrm{cov}(m,b) \tag{4}$$

is the corrector current at which the beam is centred in the corresponding quadrupole. The full covariance

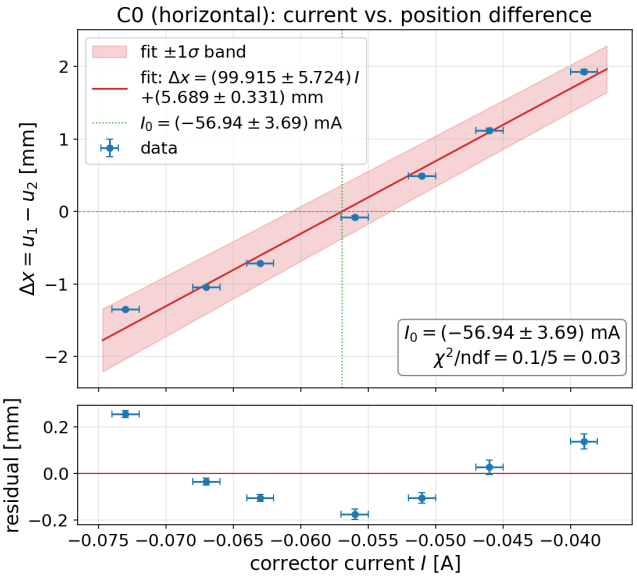


FIG. 3: Beam-based alignment of Q1 in the horizontal plane. The displacement $\Delta x = x_1 - x_2$ between the two quadrupole settings is plotted against the C0 corrector current. The zero crossing $I_0 = -57(4)$ mA defines the alignment current. The bottom panel shows the residuals of the fit.

was retained in the error propagation since m and b are strongly correlated for data not symmetric around $I = 0$. For datasets where extreme correlation between the linear parameters drives the first-order Taylor error propagation below zero, a hard floor of ±1 mA is imposed. This value represents the physical set-point resolution limit of the LAB control system power supplies. This prevents numerical artifacts from underreporting the uncertainty and ensures the error bars reflect the true instrumentation limits of the apparatus.

To convert the alignment currents I_0 to actual deflection angles, we established the relationship between corrector current and displayed kick angle by fitting $\alpha_{\mathrm{sys}} = m_\alpha \cdot I + b_\alpha$ to the same alignment data using ODR. This accounts for uncertainties in both I (0.001 A) and the displayed angle (0.1 mrad). The fitted conversion parameters m_α and b_α allow us to compute $\alpha_{\mathrm{sys}}(I_0)$ for each alignment current. The actual deflection angles are then obtained by applying the calibration factors C from Sec. III A , which correct for the systematic offset between displayed and actual angles: $\alpha_{\mathrm{actual}} = C \cdot \alpha_{\mathrm{sys}}$. Plots of the I α_{sys} fits are included in Appendix C.

A representative alignment fit is shown in Fig. 3; the plots for the remaining seven measurements are included in Appendix C. In all cases the linear model describes the trend of the data by visual inspection, with a goodness-of-fit χ^2/ndf between 0.003 and 0.31. The small reduced χ^2 values indicate that the quoted position and current uncertainties are larger than the residual scatter, so the uncertainties on I_0 are conservative.

Results

The fitted parameters and the resulting alignment currents I_0 are summarised in Tab. I. The horizontal and vertical alignment currents are of comparable magnitude but opposite sign (negative in x, positive in z for our sign convention), reflecting the misalignment of the gun in different directions in the two planes. Most measurements achieve relative uncertainties below 7 %. Q1 (z) and Q4 (x) have larger relative uncertainties of 15 % and 25 % respectively, reflecting their small alignment currents and broader data scatter compared to the other plots of Fig. 6.

TABLE I: Linear-fit parameters $\Delta u = mI + b$, reduced χ^2, and alignment currents $I_0 = -b/m$ for all four quadrupoles in both planes.

Quad.	Plane	m [mm/A]	b [mm]	χ^2/ndf	I_0 [mA]
Q1 (C0)	x	100(6)	5.7(3)	0.03	−60(4)
Q1 (C0)	z	−110(10)	6.5(8)	0.05	60(9)
Q2 (C1)	x	220(8)	12.9(5)	0.02	−59.0(1)
Q2 (C1)	z	−300(11)	16.0(6)	0.04	53.5(10)
Q3 (C2)	x	95(6)	3.6(2)	0.01	−38(1)
Q3 (C2)	z	−115(4)	11.0(3)	0.00	96(1)
Q4 (C3)	x	180(30)	3(1)	0.31	−20(5)
Q4 (C3)	z	−160(11)	14.2(11)	0.05	90(6)

After applying these currents the beam is centred in all four quadrupoles to within the quoted uncertainties. The corresponding corrector settings are documented in Tab. II, which lists the alignment currents I_0, the displayed kick angles α_{sys} at these settings, and the actual kick angles α_{actual} obtained by applying the calibration factors C from Sec. III A. For C2 and C3, which were not independently calibrated, we use the calibration factors of C1 as a best estimate, since all four corrector magnets are mechanically identical (see Sec. 2.2.2 [1]); the additional systematic uncertainty introduced by this assumption is not included in the quoted errors. However, it is important to note that because the beam-based alignment procedure determines the optimal centering current I_0 empirically, this unquantified calibration uncertainty only affects the reported physical kick angles in Table II. It is mathematically decoupled from the physical beam trajectory and therefore does not propagate into the downstream emittance measurements.

IV. EMITTANCE

A. Quadrupole Scan

We measured the horizontal and vertical emittance using the quadrupole scan technique. For each plane we performed two scans, varying the strength of either Q1 or Q3 and measuring the beam size σ at the screen S1 or

TABLE II: Final corrector settings for beam alignment. I_0 is the alignment current; $\alpha_{\rm sys}$ is the displayed kick angle at that current; $\alpha_{\rm actual} = C \cdot \alpha_{\rm sys}$ is the actual deflection angle using calibration factors from section III A. Values for C2 and C3 use the C1 calibration factors.

Corrector	Plane	I_0 [mA]	$\alpha_{\rm sys}$ [mrad]	$\alpha_{\rm actual}$ [mrad]
C0	x	$-60(4)$	$-17(1)$	$-23(2)$
C0	z	60(9)	18(3)	27(4)
C1	x	$-58.6(1)$	$-18.0(4)$	$-20.4(5)$
C1	z	53.5	16.3(4)	19.7(5)
C2	x	$-38(1)$	$-11.5(4)$	$-13.1(5)$
C2	z	96	29(1)	35(1)
C3	x	$-20(5)$	$-5(2)$	$-6(2)$
C3	z	90(6)	27.5(20)	33(2)

S3 directly downstream. The measured currents have an uncertainty of $\sigma_I = 1\,\mathrm{mA}$; the uncertainty on the beam size is taken from the framegrabber RMS of the last 10 frames.

Fit model

For each scan we fit the measured $\sigma^2(I)$ to a model derived from the linear ion-optics formalism. The transfer matrix from the entrance of the quadrupole to the screen is

$$M(I) = M_{\rm drift}(d) \cdot M_{\rm quad}(k(I)), \tag{5}$$

with $d = 0.4245\,\mathrm{m}$ the drift length from the quad exit to the screen (derived from the lattice table in Appendix A. The drift matrix is

$$M_{\rm drift}(d) = \begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}, \tag{6}$$

and the quadrupole matrix in the thick-lens form (with $L_Q = 0.074\,\mathrm{m}$ the effective field length is, in the plane in which the quadrupole focuses ($k > 0$),

$$M^{\rm foc}_{\rm quad}(k) = \begin{pmatrix} \cos\!\left(\sqrt{k}\,L_Q\right) & \dfrac{\sin\!\left(\sqrt{k}\,L_Q\right)}{\sqrt{k}} \\ -\sqrt{k}\,\sin\!\left(\sqrt{k}\,L_Q\right) & \cos\!\left(\sqrt{k}\,L_Q\right) \end{pmatrix}, \tag{7}$$

and in the defocusing plane the trigonometric functions are replaced by their hyperbolic counterparts:

$$M^{\rm def}_{\rm quad}(k) = \begin{pmatrix} \cosh\!\left(\sqrt{|k|}\,L_Q\right) & \dfrac{\sinh\!\left(\sqrt{|k|}\,L_Q\right)}{\sqrt{|k|}} \\ \sqrt{|k|}\,\sinh\!\left(\sqrt{|k|}\,L_Q\right) & \cosh\!\left(\sqrt{|k|}\,L_Q\right) \end{pmatrix}. \tag{8}$$

The quadrupole strength is computed from the current via [1]

$$k(I) = 128027 \cdot \frac{I}{p}, \qquad pc = \sqrt{T(T + 2m_e c^2)} \approx 160.4\,\mathrm{keV}, \tag{9}$$

with $T = 25\,\mathrm{keV}$ the kinetic energy. The sign convention is that $k > 0$ means the quadrupole focuses in the horizontal plane and defocuses in the vertical, and vice versa for $k < 0$. The current I can take either sign depending on the polarity of the supply. In our analysis, we therefore compute a signed $k(I)$ from the formula above (carrying the sign of I), and then for the vertical plane we use $k_{\rm eff} = -k(I)$. The sign of $k_{\rm eff}$ in each plane determines whether the focusing or the defocusing form of $M_{\rm quad}$ is the one that applies in that plane.

Carrying out the matrix product $M(I) = M_{\rm drift} \cdot M_{\rm quad}$ explicitly, the elements relevant for the fit are, in the focusing plane,

$$m_{11}(I) = \cos\!\left(\sqrt{k}\,L_Q\right) - d\sqrt{k}\,\sin\!\left(\sqrt{k}\,L_Q\right), \tag{10}$$

$$m_{12}(I) = d\,\cos\!\left(\sqrt{k}\,L_Q\right) + \frac{\sin\!\left(\sqrt{k}\,L_Q\right)}{\sqrt{k}}, \tag{11}$$

and in the defocusing plane,

$$m_{11}(I) = \cosh\!\left(\sqrt{|k|}\,L_Q\right) + d\sqrt{|k|}\,\sinh\!\left(\sqrt{|k|}\,L_Q\right), \tag{12}$$

$$m_{12}(I) = d\,\cosh\!\left(\sqrt{|k|}\,L_Q\right) + \frac{\sinh\!\left(\sqrt{|k|}\,L_Q\right)}{\sqrt{|k|}}. \tag{13}$$

Inserting these into the general expression for the propagation of the beta matrix [1]

$$\sigma_1^2 \equiv \varepsilon\beta_1 = m_{11}^2 \cdot (\varepsilon\beta_0) - 2m_{11} \cdot m_{12} \cdot (\varepsilon\alpha_0) + m_{12}^2 \cdot (\varepsilon\gamma_0) \tag{14}$$

and identifying the fit parameters as the Twiss products at the quadrupole entrance,

$$X_1 \equiv \epsilon\beta_0, \qquad X_2 \equiv \epsilon\alpha_0, \qquad X_3 \equiv \epsilon\gamma_0, \tag{15}$$

gives the fit function

$$\sigma^2(I) = m_{11}^2(I)\,X_1 - 2\,m_{11}(I)\,m_{12}(I)\,X_2 + m_{12}^2(I)\,X_3. \tag{16}$$

This is the equation we fit to the measured $\sigma^2(I)$ for each of the four data sets, with (X_1, X_2, X_3) as the three free parameters.

The fit was performed using orthogonal distance regression (ODR) with errors on both axes ($\sigma_I = 1\,\mathrm{mA}$ on the current and the framegrabber RMS on σ, propagated to σ^2 as $2\sigma\,\Delta\sigma$). The covariance matrix of the fit was carried through all subsequent computations using the `uncertainties` Python package, so all quoted uncertainties include correlations between the fit parameters.

Emittance estimators

The fit gives us the three Twiss products $(X_1, X_2, X_3) = (\epsilon\beta_0, \epsilon\alpha_0, \epsilon\gamma_0)$ at the entrance of the quadrupole, but not the emittance ϵ on its own. To isolate ϵ we used two methods.

Determinant method. The Twiss parameters at any position along the beamline are constrained by

$$\beta\gamma - \alpha^2 = 1, \tag{17}$$

which follows from their definition together with the relation $\gamma = (1 + \alpha^2)/\beta$ [1]. Multiplying both sides by ϵ^2 and inserting our fit parameters gives

$$(\epsilon\beta_0)(\epsilon\gamma_0) - (\epsilon\alpha_0)^2 = X_1X_3 - X_2^2 = \epsilon^2, \tag{18}$$

so the emittance is obtained directly from the fit parameters as

$$\epsilon = \sqrt{X_1X_3 - X_2^2}. \tag{19}$$

Waist method. At the position where the beam reaches its smallest size, the phase-space ellipse is upright, which means $\alpha_w = 0$ at the waist [1]. Combined with the constraint $\beta\gamma - \alpha^2 = 1$ from above, this gives $\gamma_w = 1/\beta_w$ at the waist. Together with the definition $\sigma_w^2 = \epsilon\beta_w$ we obtain

$$\epsilon = \frac{\sigma_w^2}{\beta_w} = \sigma_w^2 \cdot \gamma_w, \tag{20}$$

or equivalently

$$\epsilon^2 = \sigma_w^2 \cdot (\epsilon\gamma_w), \tag{21}$$

which is the form we used. Here $(\epsilon\gamma_w)$ is the value of the Twiss product at the screen position evaluated at the current I_w for which the beam is at its waist. We obtain it by transforming the beta matrix from the quadrupole entrance to the screen using [1]

$$B_{\text{screen}}(I_w) = M(I_w) \cdot B_0 \cdot M^T(I_w), \tag{22}$$

where $B_0 = \begin{pmatrix} X_1 & -X_2 \\ -X_2 & X_3 \end{pmatrix}$ is the beta matrix at the quadrupole entrance multiplied by ϵ, and the $(\epsilon\gamma_w)$ we need is the lower-right element of the transformed matrix. Giving us:

$$\begin{aligned} \epsilon^2 &= \sigma_w^2 \cdot (\epsilon\gamma_w) \\ &= \sigma_w^2 \big(m_{21}^2(I_w) \cdot X_1 - 2m_{21}(I_w)m_{22}(I_w) \cdot X_2 \\ &\quad + m_{22}^2(I_w) \cdot X_3\big) \end{aligned} \tag{23}$$

For σ_w^2 we used the smallest measured σ^2 from the data set, with its corresponding I_w, rather than the vertex of a parabolic fit. A fitted vertex would not be statistically independent of the Twiss fit, and using a measured data point keeps the two methods as decoupled as possible. The price we pay is a small bias: the measured waist is the closest sampled point to the true minimum, not the true minimum itself, so σ_w is slightly larger than its true value at the actual waist.

Even with this choice, the two methods are not fully independent since they share the same fit for the matrix propagation. The comparison between them therefore mainly serves as a consistency check on the fit, rather than as two independent measurements of the emittance.

Results

The fits and resulting emittances are shown in Figs. 4 and 8a–8c. The four data sets and their fit results are summarized in table III.

TABLE III: Emittance results from the four quadrupole scans.

Pair	Plane	χ^2/ndof	I_w [mA]	ϵ_{det} [mm mrad]	ϵ_{waist} [mm mrad]
Q1 → S1	x	0.16	30 mA	4.13(5)	4.60(7)
Q3 → S3	x	0.88	70 mA	4.58(15)	5.03(14)
Q1 → S1	z	0.40	−22 mA	3.65(7)	3.77(7)
Q3 → S3	z	0.16	−37 mA	3.34(5)	4.23(5)

a. ***Goodness of fit.*** Three of the four quadrupole scan fits exhibit a reduced chi-squared value of $\chi^2/\text{ndof} < 0.5$ (Table III). Unlike the corrector calibration, the beam centroid remains stationary on the screen during a quadrupole scan, which minimizes position-dependent spatial systematics such as phosphor inhomogeneities. In this stationary case, the temporal RMS calculated by the framegrabber provides a conservative upper bound on the uncertainty of the beam width for two reasons. Physically, because the beam does not traverse the screen, the recorded variance reflects only localized temporal fluctuations rather than genuine physical deviations from the thick-lens model. Statistically, the framegrabber reports the sample standard deviation over $N = 10$ frames. By using this raw RMS as the measurement uncertainty instead of the standard error of the mean (RMS/$\sqrt{N}$), the statistical error of the mean beam width is artificially inflated by a factor of $\sqrt{10} \approx 3.16$. This overestimation forces the χ^2 values artificially low, confirming that the residual scatter is well within the assigned precision limits and validating the thick-lens model fit.

Only the Q3 → S3 horizontal fit yields a reduced chi-squared close to unity ($\chi^2/\text{ndof} = 0.88$). For ndof = 7, this corresponds to a p-value of approximately 0.52, confirming that the residual scatter is consistent with statistical fluctuations. The remaining low χ^2/ndof values correspond to p-values approaching 1.0 (e.g., $p \approx 0.99$ for $\chi^2/\text{ndof} = 0.16$).

b. ***Comparison between the two pairs.*** The two horizontal results disagree by about 2.9σ ($\epsilon_x = 4.13$ at Q1 vs. 4.58 at Q3). The two vertical results disagree by

about 3.6σ ($\epsilon_z = 3.65$ at Q1 vs. 3.34 at Q3). Since the emittance should be conserved along the beamline (Liouville's theorem), this disagreement points to a systematic effect rather than a real difference.

For the horizontal plane, the most likely cause is the alignment quality at C3. Looking at the alignment summary in section III B , the C3-x fit has a slope uncertainty of about 20 %, while C0-x, C1-x and C2-x are all around 4–6 %. A misaligned beam through Q3 picks up an additional steering kick when the quadrupole strength is varied, which distorts the parabolic shape of $\sigma^2(I)$ and inflates the apparent emittance. The Q3 result being higher than the Q1 result is the expected sign of this effect.

For the vertical plane, the alignment of all correctors was clean, so the C3 issue does not explain the disagreement. Possible alternative causes are calibration differences between S1 and S3, residual coupling between the two planes slightly violating the assumption of completely independent planes, or systematics in the thick-lens model itself.

c. ***Comparison between the two methods.*** The waist method yields emittance values that are systematically higher than the determinant method by 3–25 %. The largest discrepancy occurs in the Q3 → S3 vertical scan (3.34 vs. 4.23 mm·mrad). While discrete sampling coarseness near the vertex might introduce a slight offset, evaluating the theoretical parabola indicates this sampling bias is negligible ($\ll 1\%$) and cannot explain the magnitude of the observed gap.

A possible explanation for this discrepancy is a systematic theoretical limitation in the waist method when applied to a fixed-screen geometry. We assume the beam waist is positioned exactly on the fluorescence screen ($\alpha_1 = 0$). However, ion-optical transport dictates that minimizing the beam spot size at a fixed longitudinal position forces the true waist to form slightly upstream of the screen. Consequently, the beam is already diverging ($\alpha_1 < 0$) upon impact. Because the Twiss parameters satisfy $\beta_1\gamma_1 = 1 + \alpha_1^2$, the waist method expression $\sqrt{\sigma_w^2(\epsilon\gamma_1)}$ mathematically evaluates to $\epsilon\sqrt{1+\alpha_1^2}$ rather than the invariant emittance ϵ. This unmodeled divergence could drive a systematic inflation of the waist method values, although a direct quantitative correction for this effect was not performed in this analysis.

Regarding precision, the statistical uncertainties yielded by both methods are of comparable magnitude across all datasets. The determinant method computes the difference of correlated variables ($X_1X_3 - X_2^2$), which can amplify the fractional error due to numerical cancellation. The waist method avoids this by evaluating a product, theoretically offering a precision advantage. However, this expected statistical gain is not observed in our results.

Given the comparable statistical precision and the likely systematic overestimation inherent to the fixed-screen waist method, we report the determinant method results as our primary values.

d. ***Final values.*** We report the Q1 → S1 determinant results as the primary emittance values for the LAB beamline:

$$\epsilon_x = 4.13(5)\,\mathrm{mm}\cdot\mathrm{mrad} \tag{24}$$

$$\epsilon_z = 3.65(7)\,\mathrm{mm}\cdot\mathrm{mrad} \tag{25}$$

The Q3 → S3 scan measurements are excluded from the final value to prevent contamination by the uncorrected C3 alignment systematic error. Averaging these results would obscure a known hardware distortion.

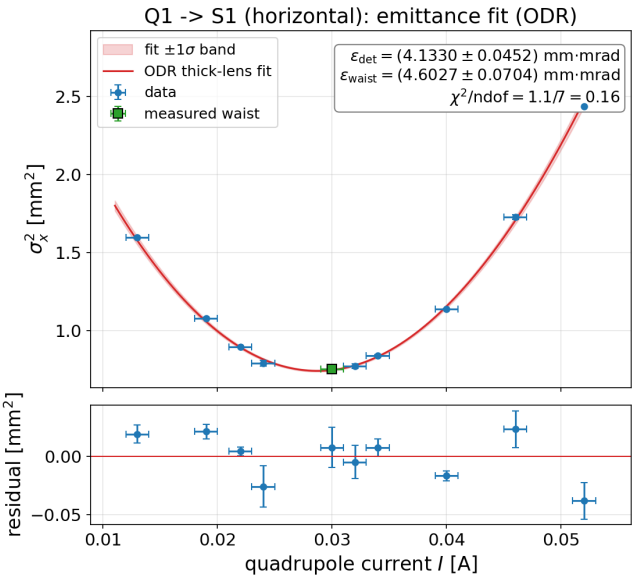


FIG. 4: Quadrupole scan Q1 → S1, horizontal plane. The thick-lens fit (red) is shown together with the $\pm1\sigma$ band, the data (blue), and the data point used as the measured waist (green square). The lower panel shows the fit residuals.

B. Multi-Screen Method

In the multi-screen method, all quadrupoles are kept at fixed currents and the beam width is measured at several screens along the beamline. From the widths, one can reconstruct the Twiss parameters at the starting point and, subsequently, the emittance.

The starting point is the transformation of the beam matrix

$$B(s_i) = M_i\, B_0\, M_i^T, \tag{26}$$

where M_i is the transfer matrix from the reference point s_0 to screen s_i. Taking the B_{11} component gives

$$\sigma_i^2 \;=\; m_{11,i}^2\,(\varepsilon\beta_0) \;-\; 2\,m_{11,i}m_{12,i}\,(\varepsilon\alpha_0) \;+\; m_{12,i}^2\,(\varepsilon\gamma_0), \tag{27}$$

with $m_{11,i}$, $m_{12,i}$ taken from the transfer matrix of the respective plane. The three combinations $\varepsilon\beta_0$, $\varepsilon\alpha_0$, $\varepsilon\gamma_0$ are the unknowns of the fit.

The transfer matrices are built separately for each plane by multiplying drift sections and quadrupole matrices in the order given in Appendix B of the script. The quadrupole strengths follow from $k = 128027 \cdot I/p$ with $p = 25\,\mathrm{keV}$ and effective length $L_{\mathrm{eff}}^{Q} = 74\,\mathrm{mm}$. A quadrupole with $k > 0$ is focusing in x and defocusing in z, so the trigonometric or hyperbolic form of the matrix is chosen accordingly. The corrector magnets are treated as drift sections since they were not powered during the measurement. The currents used are listed in Tab. IV.

TABLE IV: Quadrupole currents during the multi-screen measurement (constant for all screens).

Quadrupole	I [A]	k [m^{-2}]
Q1	0.010 A	$8\,\mathrm{m}^{-2}$
Q2	−0.006 A	$-4\,\mathrm{m}^{-2}$
Q3	−0.022 A	$-17\,\mathrm{m}^{-2}$
Q4	−0.059 A	$-47\,\mathrm{m}^{-2}$

With the four usable screens S0–S3, this provides an overdetermined linear system with four equations and three unknowns per plane. Screen S4 did not produce a fittable beam profile and is excluded. We place the reference point at $s_0 = \mathrm{S0}$, so that $M_0 = \mathbb{I}$ and the first row of the design matrix reduces to $(1, 0, 0)$. In matrix form,

$$\underbrace{\begin{pmatrix}\sigma_0^2\\ \sigma_1^2\\ \sigma_2^2\\ \sigma_3^2\end{pmatrix}}_{\vec{\sigma}} = \underbrace{\begin{pmatrix}1 & 0 & 0\\ m_{11,1}^2 & -2\,m_{11,1}m_{12,1} & m_{12,1}^2\\ m_{11,2}^2 & -2\,m_{11,2}m_{12,2} & m_{12,2}^2\\ m_{11,3}^2 & -2\,m_{11,3}m_{12,3} & m_{12,3}^2\end{pmatrix}}_{A} \underbrace{\begin{pmatrix}\varepsilon\beta_0\\ \varepsilon\alpha_0\\ \varepsilon\gamma_0\end{pmatrix}}_{\vec{x}}. \tag{28}$$

The system is solved by a weighted linear least-squares fit,

$$\hat{\vec{x}} = (A^T W A)^{-1} A^T W \vec{\sigma}, \tag{29}$$

with weights $W = \mathrm{diag}\big(1/\mathrm{Var}(\sigma_i^2)\big)$ and $\mathrm{Var}(\sigma_i^2) = (2\sigma_i \Delta\sigma_i)^2$ from standard error propagation. The covariance matrix of the fit parameters is $\Sigma_{\hat{x}} = (A^T W A)^{-1}$. The uncertainty of 1 mA on each quadrupole current propagates into the matrix elements $m_{11,i}, m_{12,i}$ and is carried through the linear-algebra operations, ensuring both the σ-measurement uncertainty and the current uncertainty appear in $\hat{\vec{x}}$ with their full correlations.

The emittance is extracted from the determinant condition of the beam matrix,

$$\varepsilon^2 = (\varepsilon\beta_0)(\varepsilon\gamma_0) - (\varepsilon\alpha_0)^2. \tag{30}$$

The uncertainty is obtained by propagating the full covariance matrix,

$$\mathrm{Var}(\varepsilon^2) = \vec{J}^T\, \Sigma_{\hat{x}}\, \vec{J}, \qquad \vec{J} = \big(\varepsilon\gamma_0,\ -2\,\varepsilon\alpha_0,\ \varepsilon\beta_0\big)^T, \tag{31}$$

and finally $\Delta\varepsilon = \Delta(\varepsilon^2)/(2\varepsilon)$.

Initial Results and Model Failure

The measured beam widths at the four screens and the resulting emittance fits using all four screens are summarised in Tab. V and Tab. VI.

TABLE V: Measured beam widths at the four screens for the multi-screen measurement.

Screen	σ_x	σ_z
S0	2.81(1) mm	3.00(1) mm
S1	1.82(1) mm	1.90(1) mm
S2	2.33(1) mm	1.87(1) mm
S3	5.10(2) mm	1.50(3) mm
S4	not fittable	not fittable

TABLE VI: Fitted Twiss products at $s_0 = \mathrm{S0}$ and the resulting emittance in each plane using all four screens.

	x	z
$\varepsilon\beta_0$	7.79(6) mm^2	8.62(7) mm^2
$\varepsilon\alpha_0$	2.5(4) µm	9.5(4) µm
$\varepsilon\gamma_0$	4.2(3)	12.9(8)
ε	5.14(10) mm mrad	4.57(11) mm mrad
χ^2/ndof	69.5/1	83.2/1

The values of χ^2/ndof are extremely high (69.5 and 83.2), proving that the physical model is wholly inconsistent with the measured beam widths given the stated statistical uncertainties. A well-behaved fit should yield $\chi^2/\mathrm{ndof} \sim 1$.

Leave-One-Out Diagnostic and Final Results

Because the reference point s_0 is located at screen S0, the initial transfer matrix M_0 is the identity matrix ($m_{11,0} = 1$ and $m_{12,0} = 0$). Consequently, the measurement at S0 provides a direct, uncoupled constraint on the initial beam size ($\sigma_0^2 = \varepsilon\beta_0$). While the downstream screens also constrain $\varepsilon\beta_0$ through their non-zero $m_{11,i}^2$ coefficients, their measurements are coupled to both $\varepsilon\alpha_0$ and $\varepsilon\gamma_0$. The presence of S0 in the fit anchors $\varepsilon\beta_0$ independently of the correlation between the other Twiss parameters.

To determine whether the massive χ^2 stems from a globally incorrect model/underestimated random errors, or a single localized systematic outlier, we performed a leave-one-out (LOO) analysis. Each screen is removed in turn and the remaining three equations are solved exactly. The results are listed in Tab. VII.

TABLE VII: Leave-one-out emittance values: each row corresponds to an exact analytical solution with one screen excluded. Quoted uncertainties are statistical only.

Dropped screen	ε_x	ε_z
none (full fit)	5.14(10) mm mrad	4.57(11) mm mrad
S0	4.5(3) mm mrad	3.9(2) mm mrad
S1	5.17(10) mm mrad	4.4(1) mm mrad
S2	5.04(10) mm mrad	3.8(1) mm mrad
S3	5.3(1) mm mrad	4.7(1) mm mrad

In both planes, dropping S1, S2, or S3 shifts the emittance by less than 0.2 mm mrad relative to the full fit. However, dropping S0 shifts the result massively by ~ 0.7 mm mrad. This demonstrates that S1, S2, and S3 are physically consistent with each other, but S0 is a systematic outlier. The beam profile back-propagated from S1, S2, and S3 does not match the directly measured σ_0^2. This discrepancy likely originates from a systematic error in the S0 measurement itself, such as severe screen degradation at that specific location or a misalignment of the reference point.

Because the error is isolated to S0 we discard the compromised S0 data point.

The most reliable multi-screen measurement is therefore the exact solution derived from screens S1, S2, and S3:

$$\varepsilon_x = 4.5(3)\,\text{mm mrad}, \tag{32}$$

$$\varepsilon_z = 3.9(2)\,\text{mm mrad}. \tag{33}$$

Comparison with the Quadrupole Scans

The multi-screen results (dropping S0) are compared with the determinant values from the quadrupole scans in Tab. VIII.

TABLE VIII: Comparison of the final multi-screen emittance (S0 excluded) and quadrupole-scan emittances (determinant method).

Method	ε_x	ε_z
Multi-screen (drop S0)	4.5(3) mm mrad	3.9(2) mm mrad
Quad-scan Q1→S1	4.13(5) mm mrad	3.65(7) mm mrad
Quad-scan Q3→S3	4.58(15) mm mrad	3.34(5) mm mrad

The multi-screen method yields emittances roughly 10% higher than the most reliable quad-scan measurements (Q1→S1). While there is still tension between the methods, it is significantly reduced compared to the full 4-screen fit. The remaining discrepancy accurately reflects unmodeled systematic factors inherent to the beamline, such as residual beam misalignment through the quadrupoles, thick-lens approximations, and unquantified screen calibration factors.

V. CONCLUSION

In this experiment, we characterized the LAB electron beamline. We performed a calibration of the corrector magnets, aligned the beam through the center of the quadrupoles, and measured the transverse emittance using the quadrupole and multi-screen method.

The kick calibration showed that the control system does not report the actual deflection angles correctly (as discussed in Sec. III A). We found calibration factors between 1.14 and 1.55, meaning the actual kicks are larger than the displayed values. These factors were essential for the following parts of the lab course. However, because we did not calibrate C2 and C3 directly, we assumed they behave like C1, which likely introduced some systematic error in the reporting of the alignment of the last two quadrupoles.

Using beam-based alignment, we were able to find the corrector currents I_0 that center the beam in each quadrupole. For this we used linear fits. The resulting alignment allowed us to vary the quadrupole strengths for the emittance measurements without accidentally steering the beam off the following quadrupole centres.

For the emittance, the quadrupole scan at Q1 provided the most consistent results. We found horizontal and vertical emittances of $\epsilon_x = 4.13(5)\,\text{mm} \cdot \text{mrad}$ and $\epsilon_z = 3.65(7)\,\text{mm} \cdot \text{mrad}$ using the determinant method. The waist method yielded systematically higher values. This discrepancy is not due to discrete sampling bias, but we think rather a theoretical limitation of the fixed-screen geometry: minimizing the spot size at a fixed location forces the true waist upstream. Consequently, the beam impacts the screen with non-zero divergence ($\alpha_1 < 0$), inflating the calculated emittance by a factor of $\sqrt{1+\alpha_1^2}$.

The multi-screen method initially yielded poor fits (χ^2/ndof values of 69.5 and 83.2). A leave-one-out diagnostic revealed that screen S0 was a severe systematic outlier, likely due to localized screen degradation or a misalignment of the reference point. Discarding S0 provided an exact analytical solution from the remaining three screens, resulting in emittance values roughly 10 % higher than the quadrupole scan measurements. This remaining discrepancy might come from unmodeled systematic factors from the beamline geometry and transport approximations.

VI. ACKNOWLEDGMENTS

We acknowledge the usage of Gemini 3.1 [2] for drafting support, text refinement, and LaTeX formatting assistance during the preparation of the whole manuscript as well as coding from Claude from Anthropic [3] throughout the entire analysis.

Appendix A: Element and Drift Lengths

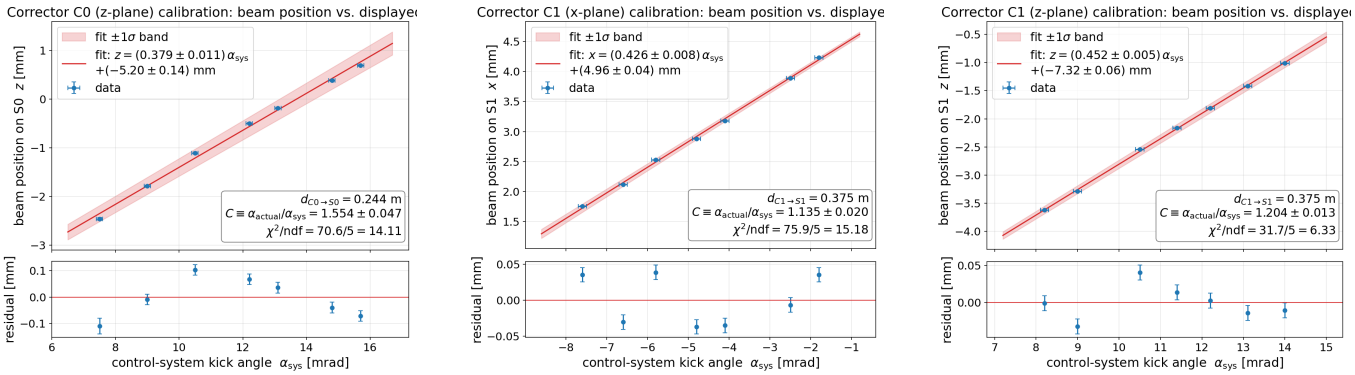
The following table lists all elements and drift spaces in the LAB beamline in their order of appearance. For the magnets, the lengths provided are the effective field lengths L_{eff}, which are essential for the ion-optical matrix calculations used throughout this report.

TABLE IX: Element types and lengths in the LAB beamline. Magnet lengths represent effective field lengths, while drift spaces indicate the free-space regions between adjacent elements.

Element	Type	Length [m]	Element	Type	Length [m]
C0	Corrector	0.0785	C2	Corrector	0.0785
	Drift	0.185		Drift	0.316
S0	Screen	0.04	S2	Screen	0.04
	Drift	0.06		Drift	0.06
Q1	Quadrupole	0.074	Q3	Quadrupole	0.074
	Drift	0.03		Drift	0.03
C1	Corrector	0.0785	C3	Corrector	0.0785
	Drift	0.316		Drift	0.316
S1	Screen	0.04	S3	Screen	0.04
	Drift	0.06		Drift	0.06
Q2	Quadrupole	0.074	Q4	Quadrupole	0.074
	Drift	0.03		Drift	0.03

Appendix B: Kick Calibration Plots

The following plots illustrate the calibration of the corrector magnets C0 and C1. By measuring the beam position on screens S0 and S1 as a function of the displayed kick angle α_{sys}, we derived the calibration factors C necessary to account for the control system's systematic underreporting of deflection angles.

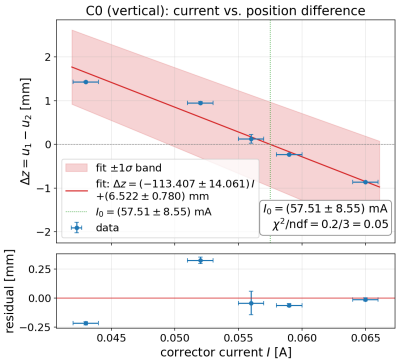


(a) C0 (z-plane) **(b)** C1 (x-plane) **(c)** C1 (z-plane)

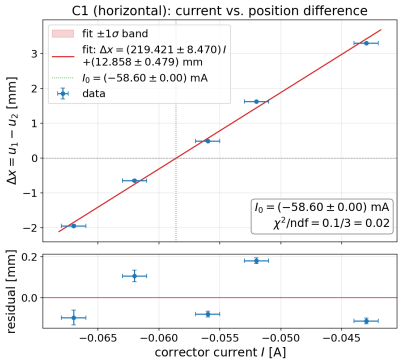
FIG. 5: Linear ODR fits for corrector kick calibration. The red lines indicate the fitted relationship used to determine the calibration factors C.

Appendix C: Beam-Based Alignment: Supplementary Measurements

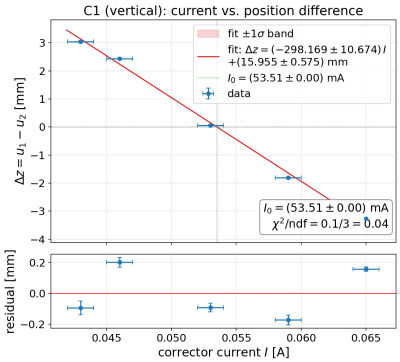
The plots shown in Fig. 6 depict the alignment process for quadrupoles Q1 through Q4. The beam is considered aligned when the position difference Δu between two different quadrupole current settings is zero, ensuring the beam passes through the magnetic center of the lens. Additionally, the current-to-angle conversion fits used to translate alignment currents into nominal system angles are shown in Fig. 7.



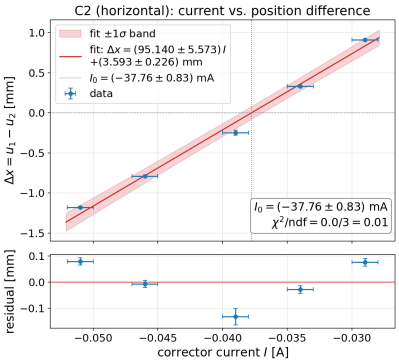
(a) Q1 Vertical (z)



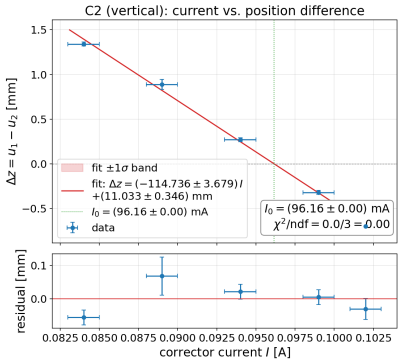
(b) Q2 Horizontal (x)



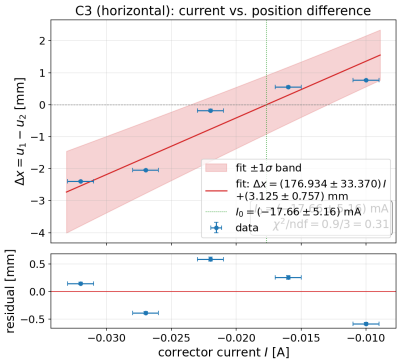
(c) Q2 Vertical (z)



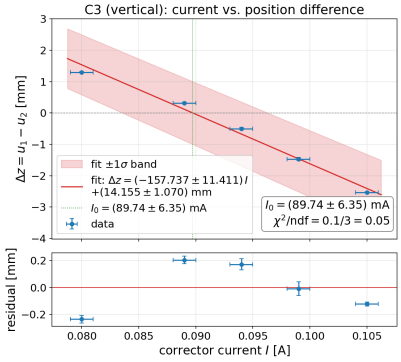
(d) Q3 Horizontal (x)



(e) Q3 Vertical (z)

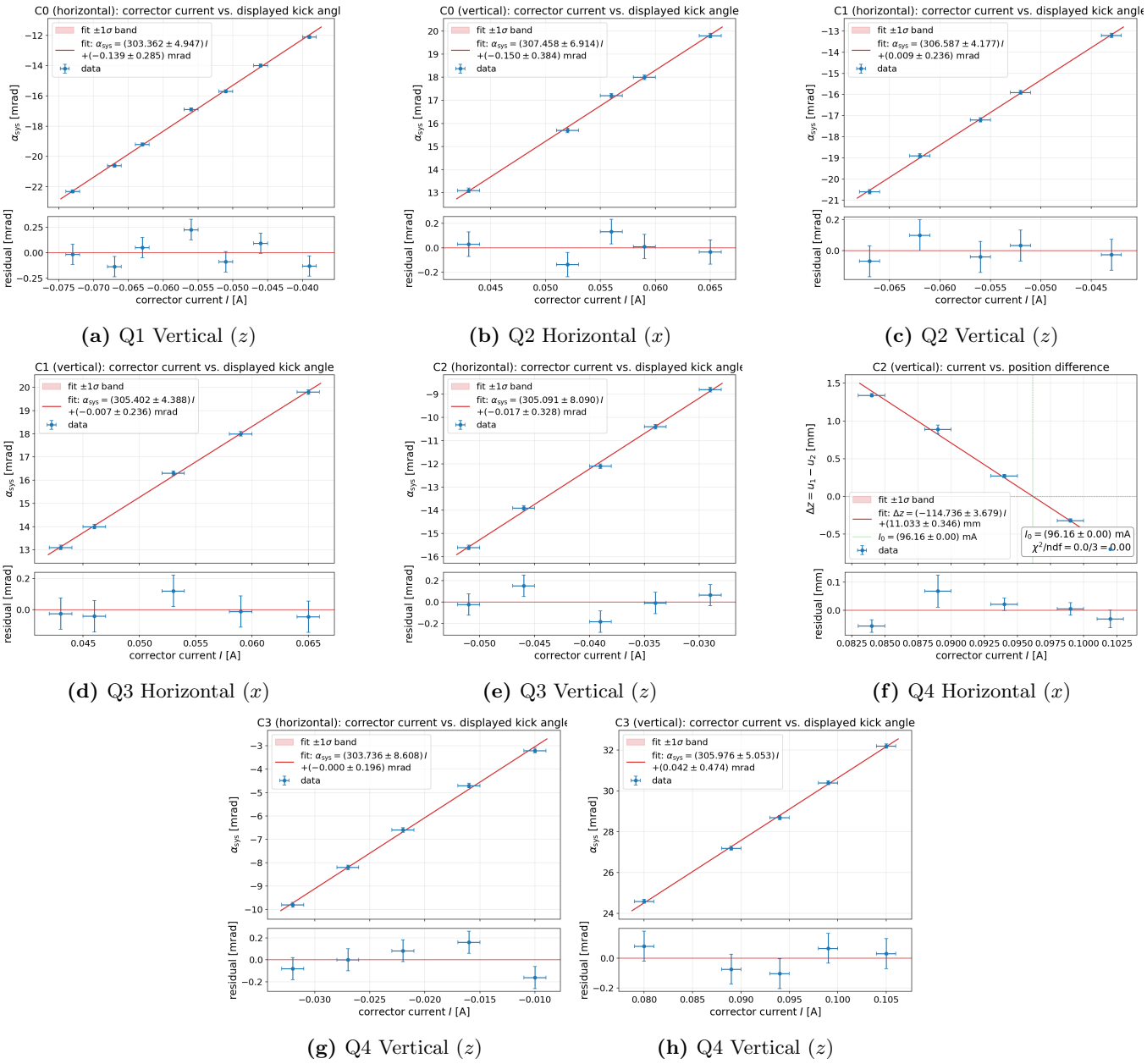


(f) Q4 Horizontal (x)



(g) Q4 Vertical (z)

FIG. 6: Beam-based alignment fits for all remaining quadrupoles. Each plot shows the displacement Δu between two quadrupole settings as a function of corrector current, with the linear fit and zero crossing I_0 indicated. Bottom panels show fit residuals.



(a) Q1 Vertical (z) **(b)** Q2 Horizontal (x) **(c)** Q2 Vertical (z)

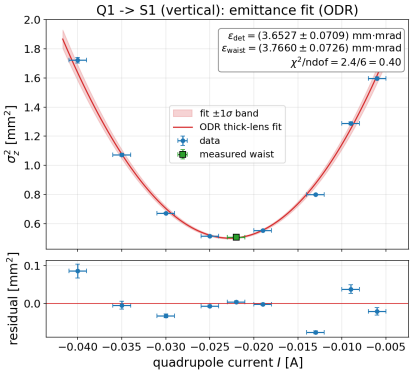
(d) Q3 Horizontal (x) **(e)** Q3 Vertical (z) **(f)** Q4 Horizontal (x)

(g) Q4 Vertical (z) **(h)** Q4 Vertical (z)

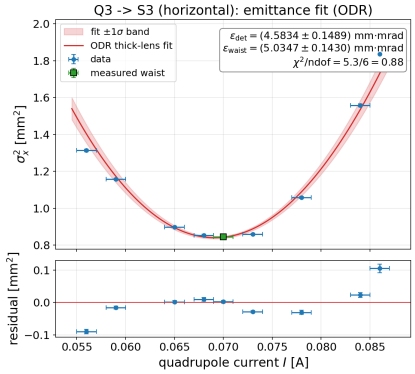
FIG. 7: Current-to-displayed-angle conversion fits for all four correctors in both planes. Each plot shows the displayed kick angle α_{sys} as a function of corrector current I, with the linear fit and confidence band. Bottom panels show fit residuals. These conversion functions are used to translate alignment currents I_0 into actual deflection angles.

Appendix D: Quadrupole Scan and Multi-Screen Fits

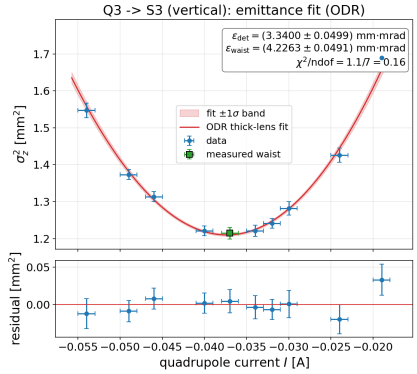
This section contains the fits for emittance determination. The quadrupole scan plots (Q1→S1 and Q3→S3) show the σ^2 evolution used in the determinant and waist methods. The multi-screen plots (Figs. 8d and 8e) display the beam envelope propagation through screens S0 to S3, with residuals highlighting the statistical tensions discussed in the report.



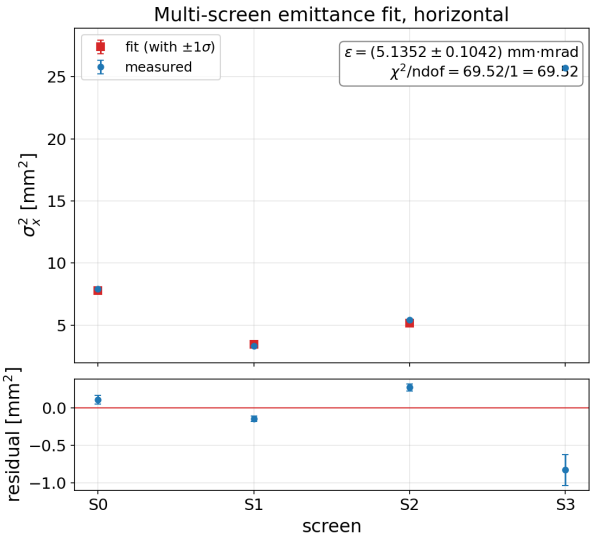
(a) Q1 → S1, Vertical Scan



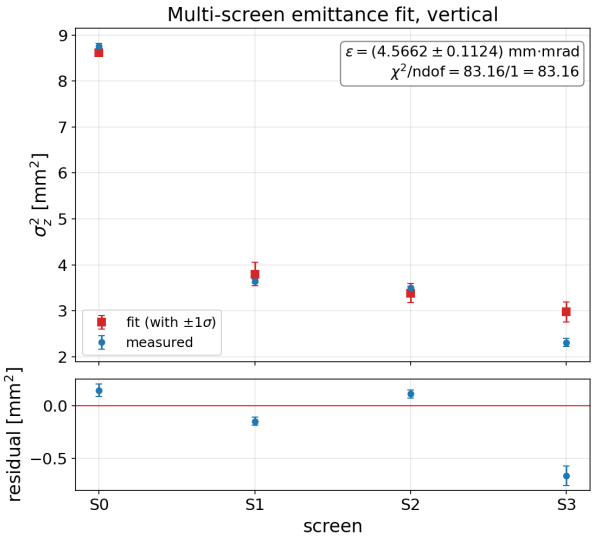
(b) Q3 → S3, Horizontal Scan



(c) Q3 → S3, Vertical Scan



(d) Multi-screen Fit, Horizontal



(e) Multi-screen Fit, Vertical

FIG. 8: Emittance measurements: comparison of scan techniques and multi-screen propagation.

[1] U. o. B. Physics Institute, Lab course accelerator bonn (lab): Script for the advanced laboratory course (physics601), experiment e207 (2025), version from July 21, 2025.

[2] Gemini Team, Google, *Gemini 3.1 Pro: Technical Report and Model Specifications*, Tech. Rep. (Google DeepMind, 2026) technical report detailing reasoning benchmarks and agentic workflows for the Gemini 3.1 series.

[3] Anthropic PBC, Anthropic — ai research and products that put safety at the frontier (2026), accessed: 2026-04-16.